

COMS4045A – Assignment

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1 Prerequisites

1.1 GitHub repository

<https://github.com/lisabeeaan/Robotics-Project>

1.2 Installations

- Fully installed ROS Kinetic and TurtleBot3 Kinetic packages.
- Verify ROS version – `rosversion -d` (should return kinetic)
- Source environments – `source /opt/ros/kinetic/setup.bash`, `source ~/yumi_ws/devel/setup.bash`
- Unpack workspace – `tar -zxvf robot_assignment_ws.tar.gz`
- Compile workspace – `rm -rf robot_assignment_ws.tar.gz`, `cd robot_assignment_ws`, `rm -rf build devel install`
- Create package – `catkin_create_pkg surveillance_bot rospy std_msgs geometry_msgs sensor_msgs nav_msgs`
- Create launch folder – `mkdir -p surveillance_bot/launch`
- Create launch file – `nano surveillance_bot/launch/gmapping.launch` (paste XML, save and exit)
- Build workspace – `cd ~/robot_assignment_ws`, `catkin_make`
- Run – `source devel/setup` (add to `.bashrc`), `./startWorld`

2 Mapping

- Install teleoperation – `sudo apt install ros-kinetic-teleop-twist-keyboard`
- Start simulation – `clear`, `./startWorldkeyboard`
- Launch SLAM – `roslaunch surveillance_bot gmapping.launch`
- Start teleoperation – `roslaunch teleop_twist_keyboard teleop_twist_keyboard.py /cmd_vel:=/cmd_vel_mux/input/teleop_twist_keyboard`
- Control robot – Use keyboard keys to navigate and map the environment.

```

ros@ASUSZenbook:~$ roslaunch teleop_twist_keyboard teleop_twist_keyboard.py
/cmd_vel:=/cmd_vel_mux/input/teleop

Reading from the keyboard and Publishing to Twist!
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Moving around:
u      i      o
j      k      l
m      ,      .

For Holonomic mode (strafing), hold down the shift key:

U      I      O
J      K      L
M      <      V

t      : up (+z)
b      : down (-z)

anything else : stop

q/z    : increase/decrease max speeds by 10%
w/x    : increase/decrease only linear speed by 10%
e/c    : increase/decrease only angular speed by 10%

CTRL-C to quit

currently:  speed 0.5      turn 1.0

```

- Save map – `roslaunch map_server map_saver -f my_map, mkdir -p ~/robot_assignment_ws/src/surveillance_bot/maps`
`cp my_map.yaml ../maps, cp my_map.pgm ../maps`

Creates:

- `my_map.pgm` → occupancy grid
- `my_map.yaml` → resolution, origin, thresholds

3 Planning

We used PRM (Probabilistic Roadmap) because the same graph can be reused for multiple navigation queries, making it more efficient than RRT. The `my_map.pgm` file is converted into a grid representation.

Learning phase – 500 random samples are generated in the occupancy map and connected to neighbouring nodes if obstacle-free. These are stored in `Samples`.

Query phase – The samples are copied, start and goal states are temporarily added and connected to neighbours, then A* computes the shortest path.

Finally, grid coordinates (x, y) are converted into world coordinates for the control stage.

4 Control

- Start simulation – `cd ~/robot_assignment_ws, ./startWorld`
- Save navigation node – `nano ../navigator_node.py` (copy code from repository)
- Give permissions – `chmod +x ../navigator_node.py`
- Create launch file – `nano ../nav.launch` (copy code from repository)
- Run navigation – `roslaunch surveillance_bot nav.launch`